National University
of Computer and Emerging Sciences
Chiniot-Faisalabad Campus

MT-1004 – Linear Algebra

Assignment # 02

Note: The last date for the submission of the assignment is November, 03, 2022

Use A4 size paper to Solve assignment

Question # 1:

If $A = \begin{bmatrix} 1 & -2 & 1 \\ -6 & 8 & -10 \\ 0 & 3 & 3 \\ -3 & 5 & -4 \end{bmatrix}$ be the matrix transformation.

a. Find order of the transformation.

b. Find a vector X whose image under A is $b = \begin{bmatrix} 1 \\ -18 \\ 9 \\ -6 \end{bmatrix}$

c. Is there more than one X whose image under A is $b = \begin{bmatrix} 1 \\ -18 \\ 9 \\ -6 \end{bmatrix}$. If possible, find three such X.

Question # 2:

If $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & 5 & -2 \end{bmatrix}$ be the matrix transformation and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_2 \end{bmatrix}$.

(a) Under what conditions on b_1, b_2 *and* b_3 , $b = \begin{bmatrix} b_1 \\ b_2 \\ b_2 \end{bmatrix}$ is in the range of A.

(b) Under what conditions on b_1, b_2 *and* b_3 , $b = \begin{bmatrix} b_1 \\ b_2 \\ b_2 \end{bmatrix}$ is not in the range of A.

Question # 3:

If $A = \begin{bmatrix} 9 & 7 & 3 & 1 \\ 1 & 2 & 3 & 5 \\ 7 & 3 & 2 & 8 \\ 3 & 6 & 4 & -2 \end{bmatrix}$ be the matrix transformation. Find all the vectors X that are mapped onto the zero vector by the transformation A.

Question # 4:

Consider a transformation defined by $T(X) = AX$, Where A is given by

$$A = \begin{bmatrix} -2 & 4 & 5 & -5 \\ 3 & -6 & -6 & 8 \\ -1 & 2 & 4 & -2 \end{bmatrix}$$

(a) What is the domain and codomain of transformation.

(b) Is $Codomain = Range$? Justify your answer mathematically.

(c) Find a vector that belongs to codomain but does not belong to range.

Question # 5:
Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be the transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle θ with clockwise rotation. Find the standard matrix of linear transformation. Also find the general formula of Linear transformation.

Question # 6:
Let $T: \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation which maps $\boldsymbol{e_1}$ onto $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\boldsymbol{e_2}$ onto $\begin{bmatrix} -5 \\ 1 \end{bmatrix}$ and $\boldsymbol{e_3}$ onto $\begin{bmatrix} 4 \\ -6 \end{bmatrix}$. Find the image of $\begin{bmatrix} -1 \\ 5 \\ 8 \end{bmatrix}$ and image of $\begin{bmatrix} x_1 \\ x_2 \\ x_2 \end{bmatrix}$.